

Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

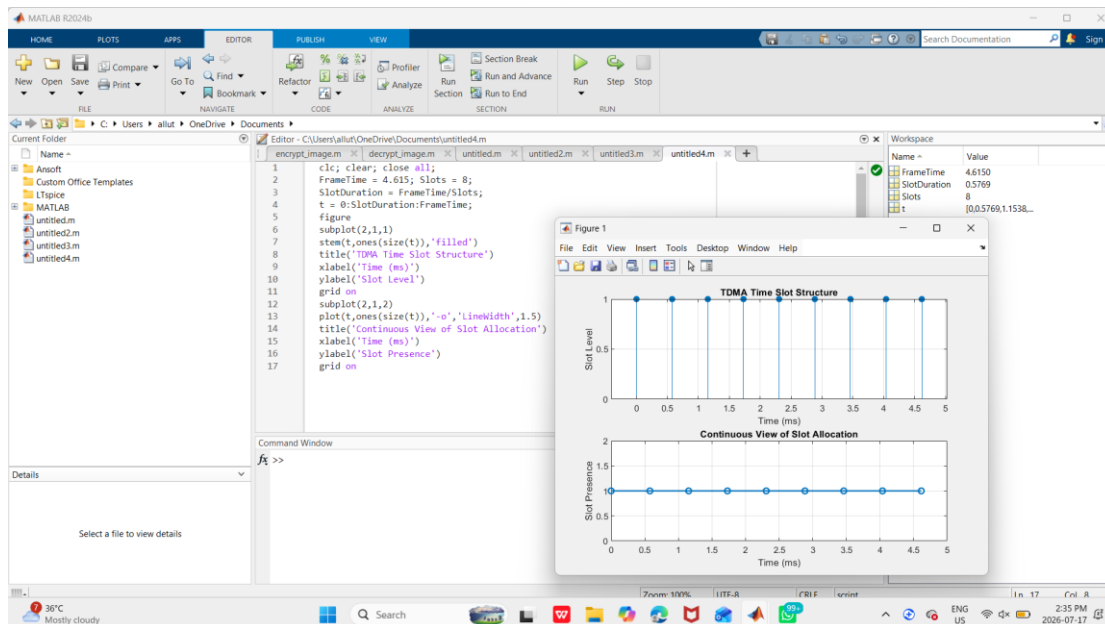
Objective:

1. To calculate the **Time Slot Duration** in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.

Objective 1: Matlab Code and Output

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

OUTPUT:



Objective 2:

To determine the **Frame Utilization (%)** based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

Matlab Code and Output :

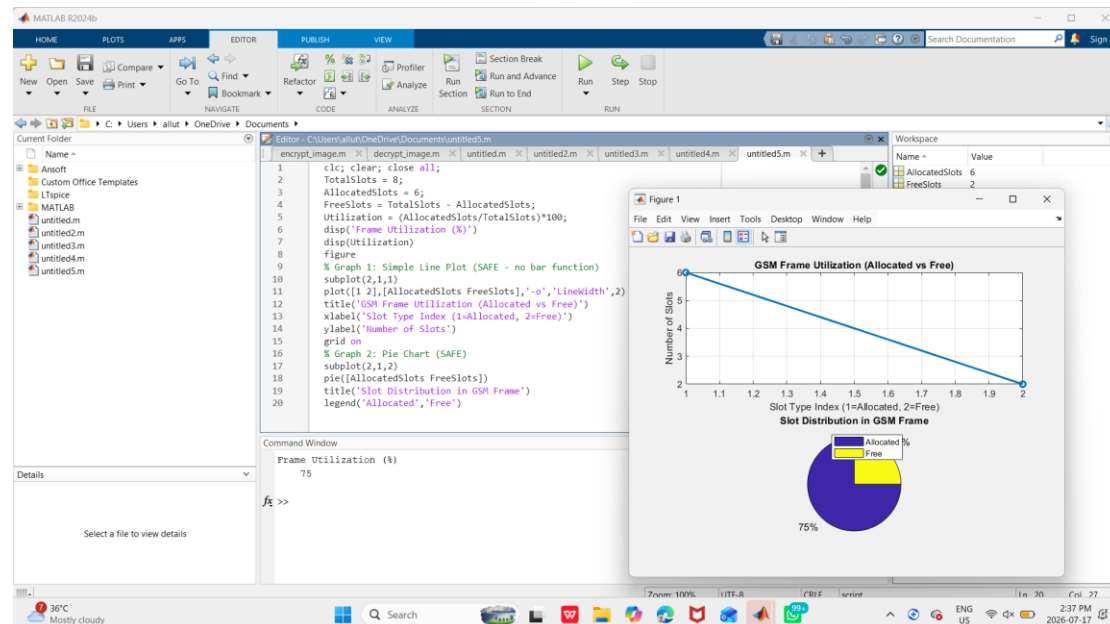
```

clc; clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on
% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')

```

legend('Allocated','Free')

OUTPUT:



Objective 3:

To compute the **Number of Allocated Time Slots** for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

Objective 3: Matlab Code and Output

```
clc; clear; close all;
Users = 1:8;
TotalSlots = 5;
Alloc = zeros(1,8);
Alloc(1:TotalSlots) = 1;
Blocked = 1 - Alloc;
disp('User Wise Slot Allocation')
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))
figure

subplot(2,1,1)stem(Users,Alloc,'filled','LineWidth',2)
```

```

title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

```

```

subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on

```

OUTPUT:

